**6** **(a)** A mass of 24 g of ice at –15 ℃ is taken from a freezer and placed in an insulated container with 200g of water at 28 ℃. Some data for the ice and the water are given in the table below.

| | specific heat capacity / J kg$^{-1}$ K$^{-1}$ | specific latent heat of fusion / J kg$^{-1}$ |
|---|---|---|
| ice | $2.1 \times 10^3$ | $3.3 \times 10^5$ |
| water | $4.2 \times 10^3$ | – |

Calculate the final temperature of the water in the container.

final temperature = ……..℃ [2]

**(b)** Some air, assumed to be an ideal gas, is contained within a cylinder. Initially, the volume of gas is 540 cm$^3$, its pressure is 1.1 x 10$^5$ Pa and its temperature is 27 ℃. The air is compress to 30 cm$^3$ suddenly such that no heat enters or leaves the gas. The pressure of the gas is 6.5 x 10$^6$ Pa immediately after the compression.

**(i)** Determine the temperature of the gas immediately after the compression.

temperature = ……..◦C [2]

**(b)** **(ii)** Explain, using the first law of thermodynamics, why the temperature of the air changes during the compression.

……...

……..

……...

……...

……...

……[3]

**(c)** The first law of thermodynamics can be expressed in the form

$$\Delta U = q + w$$

where $U$ is the internal energy of the system,
$\Delta U$ is the increase of the internal energy of the system,
$q$ is the thermal energy supplied to the system,
$w$ is the work done on the system.

The table below shows different processes. Complete the table with symbols '+' for an increase, '–' for a decrease and the symbol '0' for no change, for the respective quantities $U$, $q$ and $w$. [3]

| Process | $U$ | $q$ | $w$ |
|---|---|---|---|
| the compression of an ideal gas at constant temperature | | | |
| the heating of a solid with no expansion | | | |
| The melting of ice at 0 ◦C into water at 0 ◦C.<br><br>(Note: ice is less dense than water) | | | |